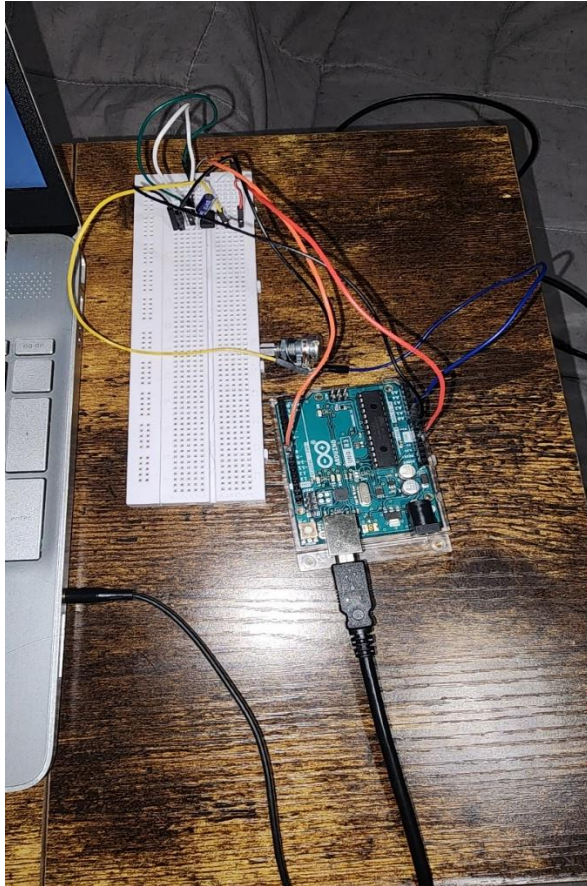


**Charlene Butler**

## **Lab Report Week 2 (90 Points)**

### **2.1 Pulse width modulation**

Paste a photo of the Arduino wired to your PWM to DC converter here. (10 points)



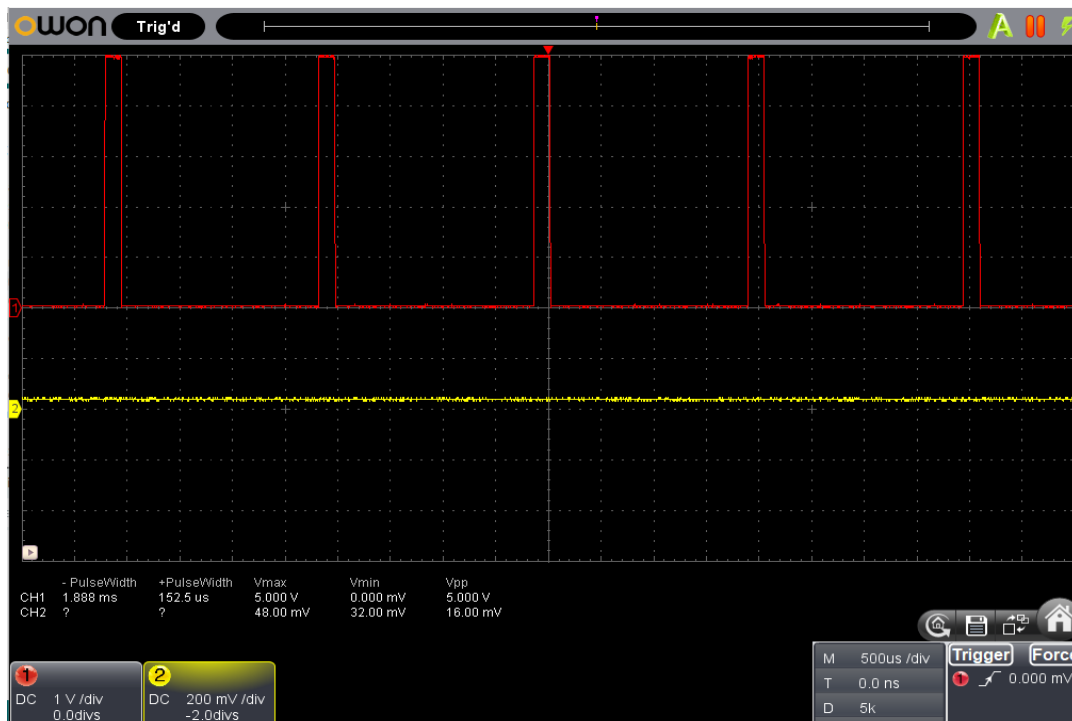
### **2.2 Quantifying PWM**

Use the Arduino program AnalogInOutSerial to generate a PWM signal from the potentiometer and convert the PWM to a DC signal using the PWM to DC converter circuit. Insert screenshots of three levels of PWM and the corresponding DC output from the op amp. Calculate the DC level from the timing of the PWM for each case and compare to your measurement (30 points).

$$Dc = (T_{on} / \text{Total Time}) * 100$$

$$\text{DC Level} = (Dc / 100) * V_{\text{supply}}$$

Smallest



Ton=152.5um

Total Time=2.04ms

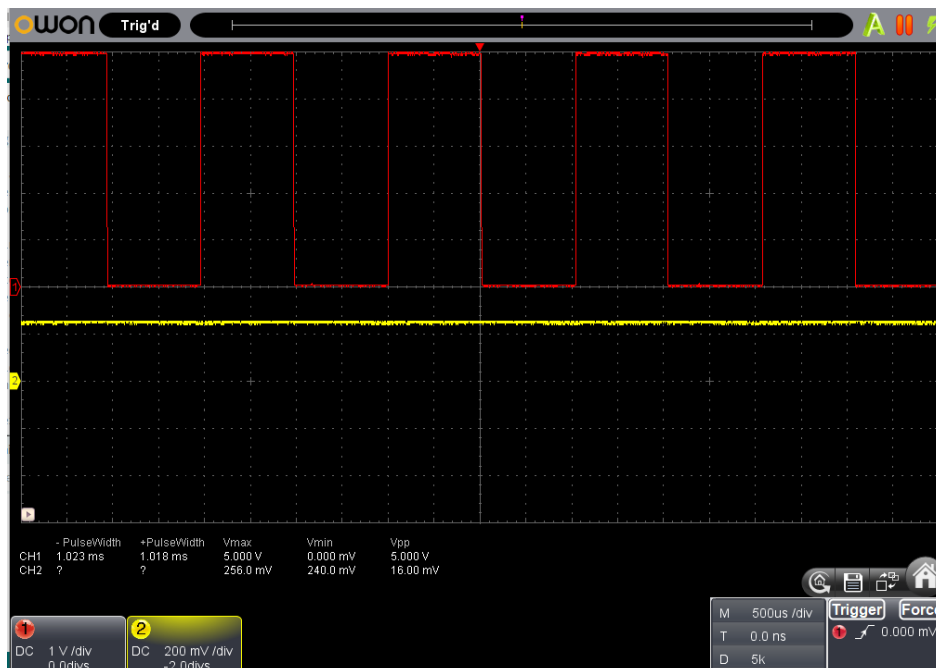
Vsupply=1V

DC measured=75mV

$$Dc = (152.5 \times 10^{-6} / 0.00204) \times 100 = 7.475$$

$$DC\ Level = (7.475 / 100) \times 1 = 0.07475V \text{ or } 74.6mV$$

Middle



Ton=1.018ms

Total Time=2.04ms

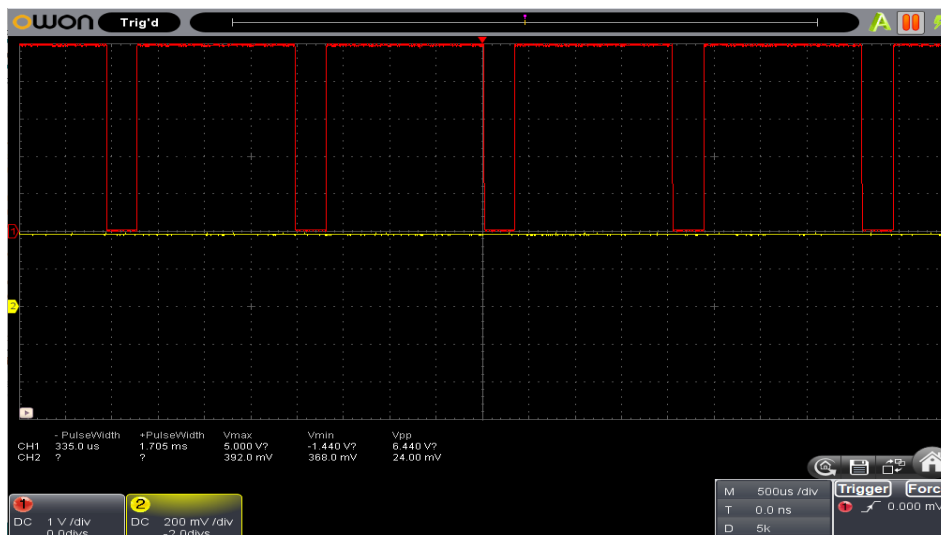
Vsupply=2.5V

DC measured=1.251V

$$Dc=(1.018*10^{-3}/0.00204)*100= 49.902$$

$$DC\ Level=(49.902/100)*2.5= 1.24755V$$

Largest



Ton=1.705ms

Total Time=2.04ms

Vsupply=4V

DC measured=3.452V

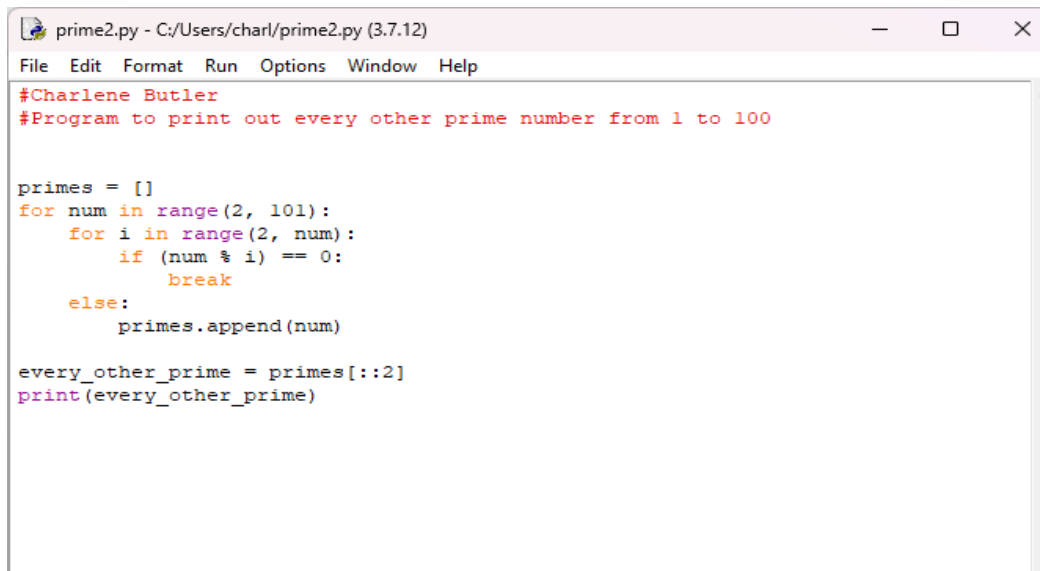
$$Dc=(1.705*10^{-3}/0.00204)*100= 83.578$$

$$DC\ Level=(83.578/100)*4= 3.343V$$

All calculations appear close to the actual measured DC levels, while the lower the supply voltage the closer the actual and measured are.

### 3.1 Python Coding

Write code to print every other prime number between 1 and 100 to your screen. Insert a screenshot of your code (20 points) and a screenshot of the output (10 points)

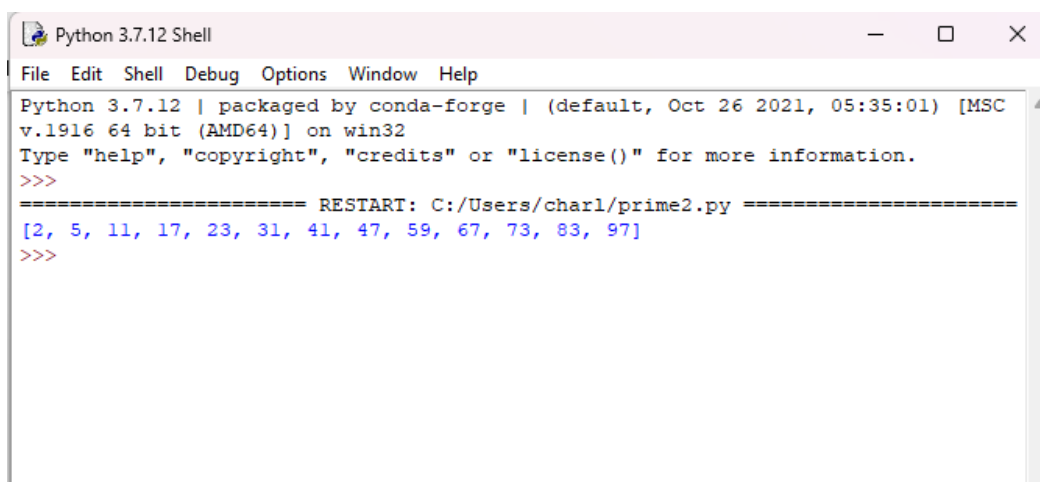


```
prime2.py - C:/Users/charl/prime2.py (3.7.12)
File Edit Format Run Options Window Help

#Charlene Butler
#Program to print out every other prime number from 1 to 100

primes = []
for num in range(2, 101):
    for i in range(2, num):
        if (num % i) == 0:
            break
        else:
            primes.append(num)

every_other_prime = primes[::2]
print(every_other_prime)
```

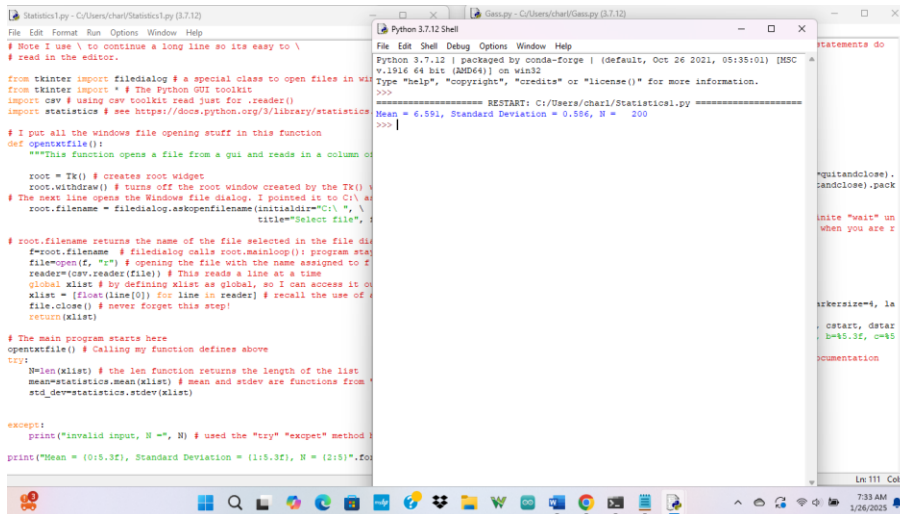


```
Python 3.7.12 Shell
File Edit Shell Debug Options Window Help

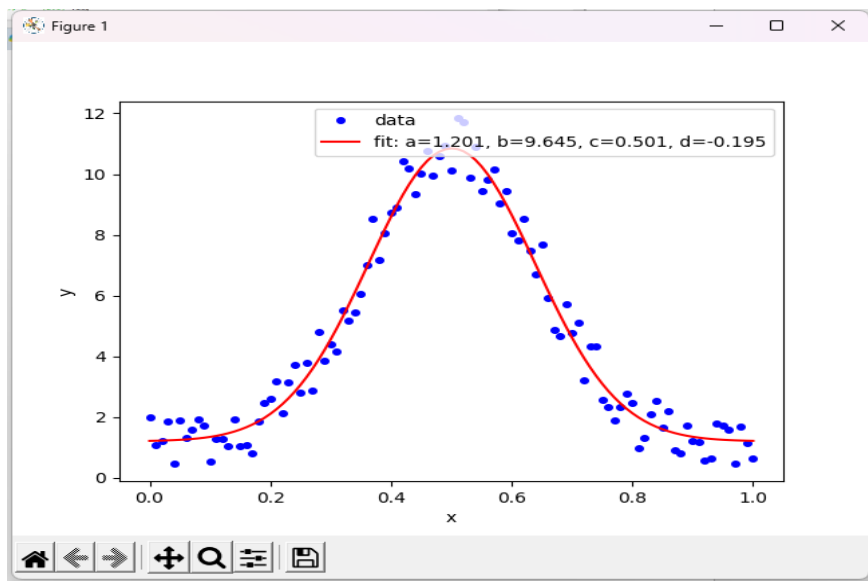
Python 3.7.12 | packaged by conda-forge | (default, Oct 26 2021, 05:35:01) [MSC
v.1916 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/charl/prime2.py =====
[2, 5, 11, 17, 23, 31, 41, 47, 59, 67, 73, 83, 97]
>>>
```

### 3.2 Using Python Programs

Submit screenshots of successful runs of Statistics.py and PlotandFitV2.py. Include both the fitted graph and the resulting covariance matrix elements (20 points)



```
Python 3.7.12 Shell
Python 3.7.12 | packaged by conda-forge | (default, Oct 26 2021, 05:35:01) [MSC
v.1916 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/charl/Statistics.py =====
>>>
Mean = 6.591, Standard Deviation = 0.556, N = 200
>>>
```



```
Python 3.7.12 Shell
File Edit Shell Debug Options Window Help

ylist = [float(line[i]) for line in reader]
IndexError: list index out of range
>>>
===== RESTART: C:/Users/charl/Statistics1.py =====
>>>
Mean = 0.500, Standard Deviation = 0.293, N = 101
>>>
===== RESTART: C:/Users/charl/Statistics1.py =====
>>>
Mean = 0.591, Standard Deviation = 0.586, N = 200
>>>
===== RESTART: C:/Users/charl/Gass2.py =====
>>>
Traceback (most recent call last):
  File "C:/Users/charl/Gass2.py", line 93, in <module>
    opensvfile()
  File "C:/Users/charl/Gass2.py", line 40, in opensvfile
    ylist = [float(line[i]) for line in reader]
  File "C:/Users/charl/Gass2.py", line 40, in <listcomp>
    ylist = [float(line[i]) for line in reader]
IndexError: list index out of range
>>>
===== RESTART: C:/Users/charl/Gass2.py =====
>>>

Warning (from warnings module):
  File "C:/Users/charl/Gass2.py", line 97
    plt.plot(xlist, ylist, 'b-', linestyle='none', marker='o', markersize=6, label='data')
UserWarning: linestyle is redundantly defined by the 'linestyle' keyword argument and the fmt string "b-" (-> linestyle=''). The keyword argument will take precedence
.

Warning (from warnings module):
  File "C:/Users/charl/Gass2.py", line 130
    plt.plot(xlist, ylist, 'b-', linestyle='none', marker='o', markersize=6, label='data')
UserWarning: linestyle is redundantly defined by the 'linestyle' keyword argument and the fmt string "b-" (-> linestyle=''). The keyword argument will take precedence
.

FINISHED FITTING
The optimized parameters are:
[ 1.20114234  5.6450532  0.50078113 -0.19531978]
The covariance matrix elements are:
[[ 1.27081628e-02 -9.04833542e-03  1.03329448e-08  3.61241144e-04]
 [-9.04833542e-03  2.02022256e-02 -7.53479468e-09  3.66378531e-05]
 [ 1.03329448e-08 -7.53479468e-09  5.90504000e-06  2.86108607e-10]
 [ 3.61241144e-04  3.66378531e-05  2.86108607e-10  2.21706489e-05]]
>>>
```